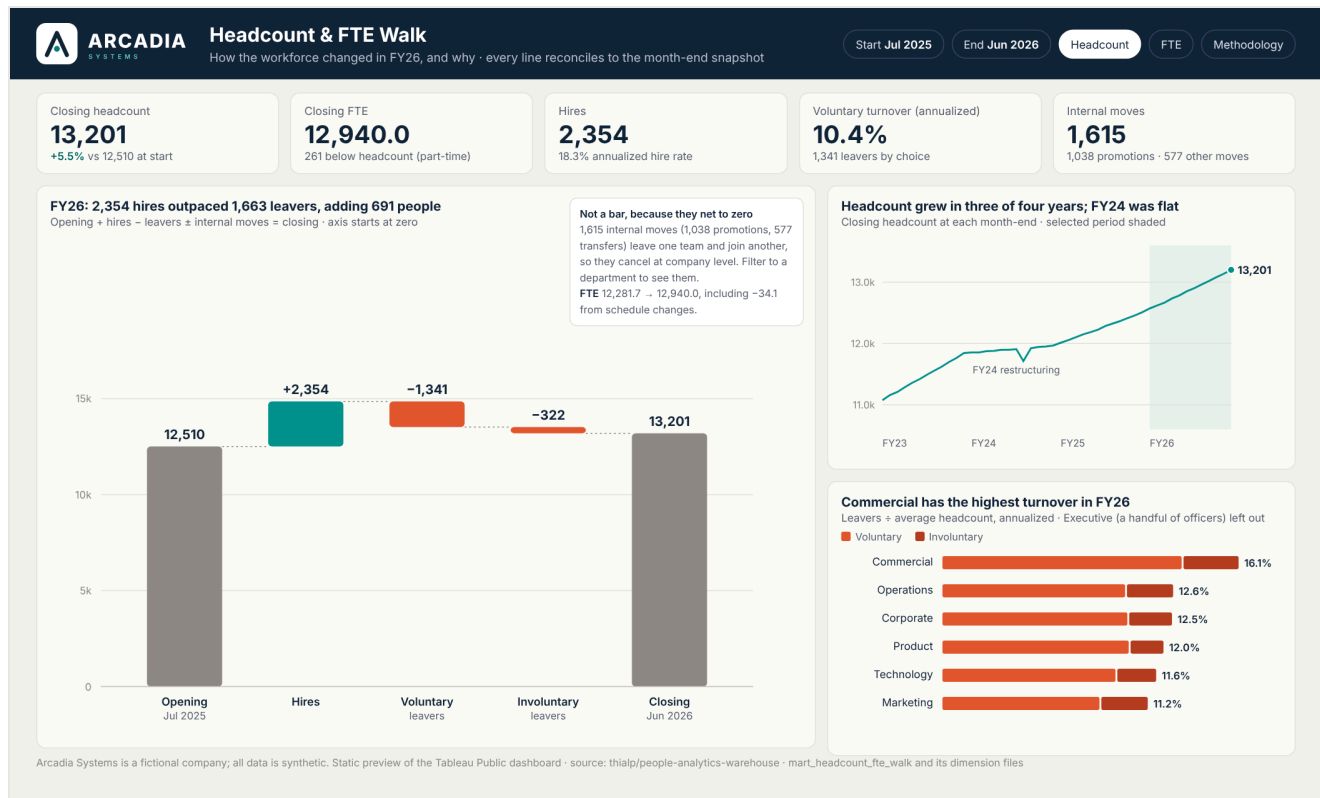


KPI cards as map layers: the Headcount & FTE Walk card band



The five cards under the header (Closing headcount, Closing FTE, Hires, Voluntary turnover, Internal moves) are **one Tableau sheet**. Every title, number and note on it is text drawn as a map layer, positioned by calculated fields. A transparent card image sits underneath. The result:

- Five cards, 17 text pieces, one sheet, one query.
- Numbers, notes and colors update with the Start Month, End Month and any filter.
- The layout (card width, gaps, text positions) is controlled by 16 parameters, so you can restyle the band without redrawing anything.
- No images carry numbers, so nothing is stale and everything is searchable and accessible as a sheet.

How it works in one paragraph. A map sheet places marks by latitude and longitude, and Tableau can draw a text label on any mark. If we treat the sheet's pixels as a tiny flat map, a pixel position becomes a (latitude, longitude) pair, a label drawn at that point appears at that pixel, and a calculated field can compute the position from parameters. Each text piece is its own map layer (a Circle mark with 0% opacity, so only its label shows).

Tested? The SQL, the numbers and the string formulas in this guide were checked against the warehouse (see Section 9). I could not run Tableau Public here, so the layer settings are

written from how the technique works. Section 6 (calibration) is where you tune it on your screen; budget 15 minutes for it.

Download the companion workbook [Arcadia_KPI_Cards_Config.xlsx](#): every parameter, calculation, layer and dashboard setting in this guide as copy-ready tables, plus a validation tab with the FY26 numbers.

1. What you need first

- The Headcount & FTE Walk data source and the calculated fields from [that guide](#), Section 4: **In Range**, **Hires**, **Voluntary Leavers**, **Voluntary Turnover (annualized)**, **Hire Rate (annualized)**, **Internal Moves**, **Average Headcount**, **Months in Range**, and the parameters **Start Month** and **End Month**.
- The card background image [brand/arcadia_kpi_cards_bg.png](#): a transparent 1400 × 110 picture (drawn at 2×) holding five rounded off-white cards with a 1px #E4E3DD border.
- A new, empty worksheet named **KPI Cards**.

2. Layout in pixels

The dashboard is 1400 × 850. The header band is 0–76. The KPI sheet floats at **x = 0, y = 78, width 1400, height 110**, and the background image floats at the same position and size, **underneath** it. The whole dashboard uses a 56 px side gutter, so cards and charts line up.

Item	Value (px)
Sheet size	1400 × 110
Left gutter (first card's left edge)	56
Card top inside the sheet	12
Card size	244 × 86
Gap between cards	17 (card lefts at 56, 317, 578, 839, 1100)
Text inset from the card's left edge	16
Title / value / note baselines below the card top	19 / 44 / 68

Why the 56 px gutter: Tableau pads the map's fitted extent by roughly 4.5% on each side, so text anchored in the outer 9% of the sheet drifts off. The first text anchor sits at 72 px, which stays inside.

3. Parameters

Create these 16 parameters (Float unless noted; none are shown on the dashboard). The prefix `kpi_` keeps them together in the Parameters list.

Name	Default	What it does
kpi_Scale	0.010986328125	Degrees per pixel at zoom 7: $360 \div (256 \times 2^7)$. Don't change it.
kpi_SheetW / kpi_SheetH	1400 / 110	The sheet's pixel size
kpi_FitX / kpi_FitY	1.1 / 1.1	Calibration factors (Section 6)
kpi_ShiftX / kpi_ShiftY	0 / 0	Nudge everything by pixels
kpi_GridLeft	56	Left edge of card 1
kpi_GridTop	12	Top edge of the cards
kpi_CardW / kpi_CardH	244 / 86	Card size
kpi_GapX	17	Space between cards
kpi_PadX	16	Text inset from the card's left edge
kpi_TitleDY / kpi_ValueDY / kpi_SubDY	19 / 44 / 68	Text center below the card top

4. Calculated fields

4.1 The numbers (prefix K)

These reuse your existing fields. Create them in order.

```
// K Valid (a start after the end is meaningless)
[Start Month] <= [End Month]
```

```
// K Open HC
ZN(SUM(IF [In Range] AND [Movement Category] = "Opening" AND [Month End Date] = [Start
Month]
      THEN [Headcount] END))
```

```
// K Close HC
ZN(SUM(IF [In Range] AND [Movement Category] = "Closing" AND [Month End Date] = [End
Month]
      THEN [Headcount] END))
```

```
// K Close FTE
ZN(SUM(IF [In Range] AND [Movement Category] = "Closing" AND [Month End Date] = [End
Month]
      THEN [Fte] END))
```

```
// K Net Pct
IF [K Open HC] = 0 THEN 0 ELSE ([K Close HC] - [K Open HC]) / [K Open HC] END
```

```
// K Hires  
ZN([Hires])
```

```
// K Leavers Vol  
ZN([Voluntary Leavers])
```

```
// K Moves  
ZN([Internal Moves])
```

```
// K Promotions  
ZN(SUM(IF [In Range] AND [Movement Category] = "Internal Moves In"  
        AND [Movement Reason] = "Promotion" THEN [Headcount] END))
```

```
// K Other Moves  
[K Moves] - [K Promotions]
```

```
// K Part Time Gap (headcount minus FTE)  
[K Close HC] - [K Close FTE]
```

Cards 1 and 2 always show **headcount** and **FTE** by name, so they ignore the Measure parameter; that parameter drives the waterfall.

4.2 Text helpers

Tableau has no number-format function for text, so we build the strings with whole-number math. DIV is integer division and % is the remainder.

```
// K Open Str (12510 -> "12,510")  
IF [K Open HC] >= 1000  
THEN STR(DIV(INT([K Open HC]), 1000)) + "," + RIGHT("00" + STR(INT([K Open HC]) % 1000),  
3)  
ELSE STR(INT([K Open HC])) END
```

```
// K Net Pct Str (0.055 -> "5.5%"; sign handled by the note)  
STR(DIV(INT(ROUND(ABS([K Net Pct]) * 1000, 0)), 10)) + "." +  
STR(INT(ROUND(ABS([K Net Pct]) * 1000, 0)) % 10) + "%"
```

Every other string repeats one of these two patterns: the **comma pattern** (for counts) and the **percent pattern** (for rates). Replace the field name in each.

4.3 What each card shows

Card	Value field	Note field(s)
1 Closing headcount	KPI 1 Value	KPI 1 Note Up, Note Down, Note Flat
2 Closing FTE	KPI 2 Value	KPI 2 Note
3 Hires	KPI 3 Value	KPI 3 Note
4 Voluntary turnover (annualized)	KPI 4 Value	KPI 4 Note
5 Internal moves	KPI 5 Value	KPI 5 Note

```
// KPI 1 Value
IF NOT [K Valid] THEN "-"
ELSEIF [K Close HC] >= 1000
THEN STR(DIV(INT([K Close HC]), 1000)) + "," + RIGHT("00" + STR(INT([K Close HC]) % 1000), 3)
ELSE STR(INT([K Close HC])) END
```

```
// KPI 1 Note Up    (teal text)
IF [K Valid] AND [K Net Pct] > 0 THEN "+" + [K Net Pct Str] + " vs " + [K Open Str] + "
at start" END
```

```
// KPI 1 Note Down  (dark coral text)
IF [K Valid] AND [K Net Pct] < 0 THEN "-" + [K Net Pct Str] + " vs " + [K Open Str] + "
at start" END
```

```
// KPI 1 Note Flat   (slate text)
IF [K Valid] AND [K Net Pct] = 0 THEN [K Net Pct Str] + " vs " + [K Open Str] + "
at start" END
```

The three notes sit on the same spot; only one is ever non-empty, so the color follows the sign without any color field.

```
// KPI 2 Value    (12,940.0: round to tenths, then split whole and tenth)
IF NOT [K Valid] THEN "-" ELSE
  IF DIV(INT(ROUND([K Close FTE] * 10, 0)), 10) >= 1000
  THEN STR(DIV(DIV(INT(ROUND([K Close FTE] * 10, 0)), 10), 1000)) + "," +
    RIGHT("00" + STR(DIV(INT(ROUND([K Close FTE] * 10, 0)), 10) % 1000), 3)
  ELSE STR(DIV(INT(ROUND([K Close FTE] * 10, 0)), 10)) END
  + "." + STR(INT(ROUND([K Close FTE] * 10, 0)) % 10)
END
```

```
// KPI 2 Note
IF NOT [K Valid] THEN "" ELSE
  IF INT(ROUND([K Part Time Gap], 0)) >= 1000
  THEN STR(DIV(INT(ROUND([K Part Time Gap], 0)), 1000)) + "," +
    RIGHT("00" + STR(INT(ROUND([K Part Time Gap], 0)) % 1000), 3)
  ELSE STR(INT(ROUND([K Part Time Gap], 0))) END
  + " below headcount (part-time)"
END
```

```
// KPI 3 Value (the comma pattern on [K Hires])
IF NOT [K Valid] THEN "-"
ELSEIF [K Hires] >= 1000
THEN STR(DIV(INT([K Hires]), 1000)) + "," + RIGHT("00" + STR(INT([K Hires]) % 1000), 3)
ELSE STR(INT([K Hires])) END
```

```
// KPI 3 Note (the percent pattern on the annualized hire rate)
IF NOT [K Valid] THEN "" ELSE
  STR(DIV(INT(ROUND(ZN([Hire Rate (annualized)]) * 1000, 0)), 10)) + "." +
  STR(INT(ROUND(ZN([Hire Rate (annualized)]) * 1000, 0)) % 10) + "% annualized hire rate"
END
```

```
// KPI 4 Value (hidden for groups under 20 people, like Rate Shown)
IF NOT [K Valid] OR [Average Headcount] < 20 THEN "-" ELSE
  STR(DIV(INT(ROUND([Voluntary Turnover (annualized)] * 1000, 0)), 10)) + "." +
  STR(INT(ROUND([Voluntary Turnover (annualized)] * 1000, 0)) % 10) + "%"
END
```

```
// KPI 4 Note (the comma pattern on [K Leavers Vol])
IF NOT [K Valid] THEN "" ELSE
  IF [K Leavers Vol] >= 1000
  THEN STR(DIV(INT([K Leavers Vol]), 1000)) + "," + RIGHT("00" + STR(INT([K Leavers Vol])
% 1000), 3)
  ELSE STR(INT([K Leavers Vol])) END
  + " leavers by choice"
END
```

```
// KPI 5 Value (the comma pattern on [K Moves])
IF NOT [K Valid] THEN "-"
ELSEIF [K Moves] >= 1000
THEN STR(DIV(INT([K Moves]), 1000)) + "," + RIGHT("00" + STR(INT([K Moves]) % 1000), 3)
ELSE STR(INT([K Moves])) END
```

```
// KPI 5 Note
IF NOT [K Valid] THEN "" ELSE
  IF [K Promotions] >= 1000
  THEN STR(DIV(INT([K Promotions]), 1000)) + "," + RIGHT("00" + STR(INT([K Promotions]) %
1000), 3)
  ELSE STR(INT([K Promotions])) END
  + " promotions · " +
  IF [K Other Moves] >= 1000
  THEN STR(DIV(INT([K Other Moves]), 1000)) + "," + RIGHT("00" + STR(INT([K Other Moves])
% 1000), 3)
  ELSE STR(INT([K Other Moves])) END
  + " other moves"
END
```

The titles are not fields: type them directly into each title layer's label (Section 5).

4.4 Geometry

Pixel to degrees. With the origin at the sheet center, longitude is the x offset and latitude is the **negative** y offset (pixels grow downward, latitude grows upward):

$$\text{lon} = (x - W/2) \times \text{scale} \times \text{FitX} \quad \text{lat} = -(y - H/2) \times \text{scale} \times \text{FitY}$$

MAKEPOINT takes **(latitude, longitude)**, in that order.

Three latitude fields (one per text row):

```
// kpi_lat Title
-([kpi_GridTop] + [kpi_TitleDY] + [kpi_ShiftY] - [kpi_SheetH] / 2) * [kpi_Scale] *
[kpi_FitY]
```

```
// kpi_lat Value
-([kpi_GridTop] + [kpi_ValueDY] + [kpi_ShiftY] - [kpi_SheetH] / 2) * [kpi_Scale] *
[kpi_FitY]
```

```
// kpi_lat Sub
-([kpi_GridTop] + [kpi_SubDY] + [kpi_ShiftY] - [kpi_SheetH] / 2) * [kpi_Scale] *
[kpi_FitY]
```

Five longitude fields (*i* is 0 to 4; write the number out in each field):

```
// kpi_lon C1 (i = 0; C2 uses 1, C3 uses 2, C4 uses 3, C5 uses 4)
([kpi_GridLeft] + 0 * ([kpi_CardW] + [kpi_GapX]) + [kpi_PadX] + [kpi_ShiftX] -
[kpi_SheetW] / 2)
* [kpi_Scale] * [kpi_FitX]
```

Fifteen anchor points, one per title, value and note on each card (names kpi_pt Title 1 ... kpi_pt Sub 5):

```
// kpi_pt Title 1
MAKEPOINT([kpi_lat Title], [kpi_lon C1])
```

Title *n* uses kpi_lat Title and kpi_lon C*n*; Value *n* uses kpi_lat Value; Note *n* (called kpi_pt Sub *n*) uses kpi_lat Sub.

Two helper lines:

```
// kpi_Frame (not Fit-scaled: it marks the sheet's true extent; drawn at 0% opacity)
MAKELINE(MAKEPOINT([kpi_SheetH] / 2 * [kpi_Scale], -[kpi_SheetW] / 2 * [kpi_Scale]),
MAKEPOINT(-[kpi_SheetH] / 2 * [kpi_Scale], [kpi_SheetW] / 2 * [kpi_Scale]))
```

```
// kpi_Check (card 1 bottom-left to card 5 top-left, Fit applied; used only to
calibrate)
MAKELINE(
MAKEPOINT(-([kpi_GridTop] + [kpi_CardH] - [kpi_SheetH] / 2) * [kpi_Scale] * [kpi_FitY],
([kpi_GridLeft] - [kpi_SheetW] / 2) * [kpi_Scale] * [kpi_FitX]),
MAKEPOINT(-([kpi_GridTop] - [kpi_SheetH] / 2) * [kpi_Scale] * [kpi_FitY],
([kpi_GridLeft] + 4 * ([kpi_CardW] + [kpi_GapX]) - [kpi_SheetW] / 2) *
[kpi_Scale] * [kpi_FitX]))
```

There is **no gate** on the points: every row produces the same point, so Tableau draws one mark per layer, and the label measures aggregate every row. The one rule: **no other dimension may be on a layer's Marks card**, or Tableau splits the layer into one mark per value.

5. Build the sheet

1. In **KPI Cards**, double-click `kpi_Frame`. Tableau creates a map. This is the bottom layer. In the Marks card set **Color opacity to 0%**, so it is invisible.
2. Hide everything map-like: **Map → Background Maps → None; Map → Map Options:** untick *Show Map Search*, *Show View Toolbar* and *Allow Pan/Zoom*. Set the sheet to **Entire View** and tooltips to none (**Worksheet → Tooltip**, untick *Show tooltips*).
3. Drag `kpi_pt Title 1` onto the map and drop it on **Add a Marks Layer**. Then in that layer's Marks card:
 - Mark type **Circle**, size at the **smallest**, Color opacity **0%**.
 - Drag **no other field** except the label text.
 - **Label → Text** (click the `...` button): for a title, type the text directly; for a value or note, insert the field.
 - Label font, alignment **Left**, vertical **Middle**, *Allow labels to overlap other marks* on.
4. Repeat for all 17 layers (15 anchor points, plus the two extra notes for card 1's Down and Flat states, which reuse `kpi_pt Sub 1`). Fonts:

Layer	Font
Titles	Tableau Semibold, 8 pt, slate #5A6170
Values	Tableau Bold, 20 pt, navy #13233A
Notes	Tableau Book, 8 pt, slate #5A6170; for KPI 1 Note Up use teal text #006B66, for KPI 1 Note Down dark coral #B8401F

Tableau Public embeds only Tableau's own fonts, so stay with these. 5. Last, add `kpi_Check` as the top layer: Mark type Line, color magenta, width 2 (you will delete it after calibration). 6. Hide the "null" indicator in the corner (right-click it → **Hide Indicator**).

6. Calibrate (about 15 minutes)

1. Make the dashboard (Section 7) with the background PNG underneath, and the sheet on top.
2. Look at the magenta check line: its two ends should land on the **bottom-left corner of card 1** and the **top-left corner of card 5**.
3. If the line is too short or too long horizontally, change `kpi_FitX`; vertically, `kpi_FitY`. Change both in steps of 0.01. They should end up close to 1.1 (Tableau pads the fit by about 4.5% on each side).
4. If the line is correct but a text piece looks off by a few pixels, change `kpi_ShiftX/kpi_ShiftY` or the text row offsets (`kpi_TitleDY`, `kpi_ValueDY`, `kpi_SubDY`).
5. If a title is centered on its anchor instead of starting there, set the label alignment to **Left** in that layer.
6. Delete the `kpi_Check` layer.

7. Put it on the dashboard

1. Dashboard size: **Fixed, 1400 × 850**.
2. Drag an **Image** object, set it to **Floating**, choose `docs/brand/arcadia_kpi_cards_bg.png`, position **x 0, y 78, w 1400, h 110**. In its settings, tick **Fit Image** and leave **Center Image** off.
3. Drag the **KPI Cards** sheet, also **Floating**, at the same position and size, and move it above the image in the layout tree. Set its background to **None** and hide its title.
4. Check that no other object overlaps this band, and that every other card on the page starts at the same left gutter, `x = 56`.

8. Header controls

The header buttons (Start, End, Headcount, FTE, Methodology) do **not** need an image per state. See the *Parameter action* entry in [the headcount guide, Section 6](#): one small sheet colors the selected option from the parameter itself, so the selected state follows automatically.

9. Check values (FY26)

Start Month 2025-07-31, End Month 2026-06-30:

Card	Value	Note
1 Closing headcount	13,201	+5.5% vs 12,510 at start
2 Closing FTE	12,940.0	261 below headcount (part-time)
3 Hires	2,354	18.3% annualized hire rate
4 Voluntary turnover (annualized)	10.4%	1,341 leavers by choice
5 Internal moves	1,615	1,038 promotions · 577 other moves

I ran the same text formulas in Python against the warehouse numbers and they reproduce this table exactly. Set Start Month after End Month: all five values should show "—".

10. If something looks wrong

Symptom	Likely cause
A label is blank	A dimension is on that layer's Marks card, splitting the layer so the label sees a partial row set. Remove it.
Two copies of a label	Same cause.
The whole text block sits too low or too high	Adjust <code>kpi_ShiftY</code> , or the Fit values (Section 6).
Text clipped at the right of card 5	The text is wider than the card; shorten the note or reduce the font.
Layout changes when the dashboard is not fixed size	Use a fixed 1400 × 850 size.
Fonts look different on Tableau Public	Only Tableau's fonts are embedded; avoid others.